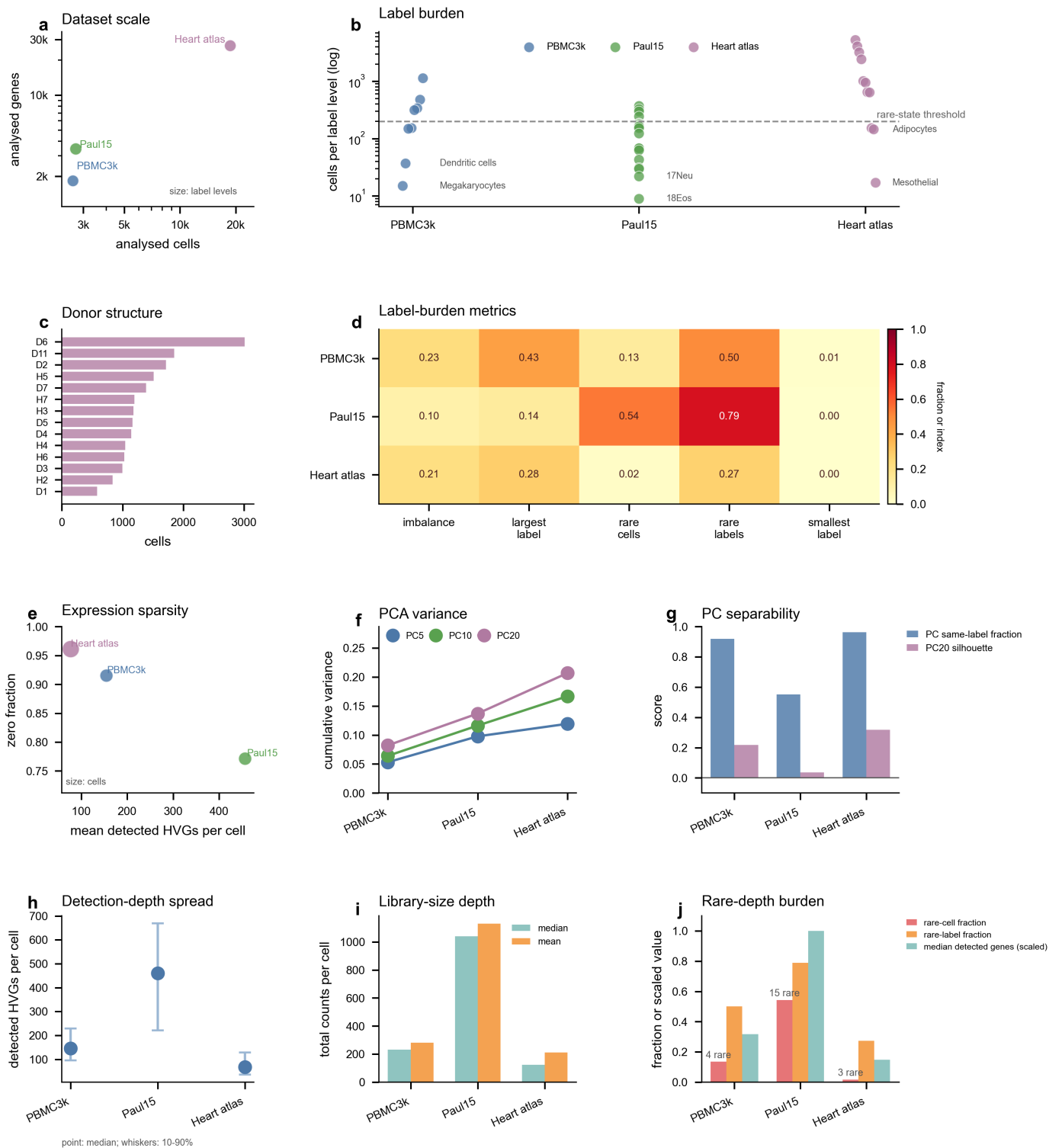
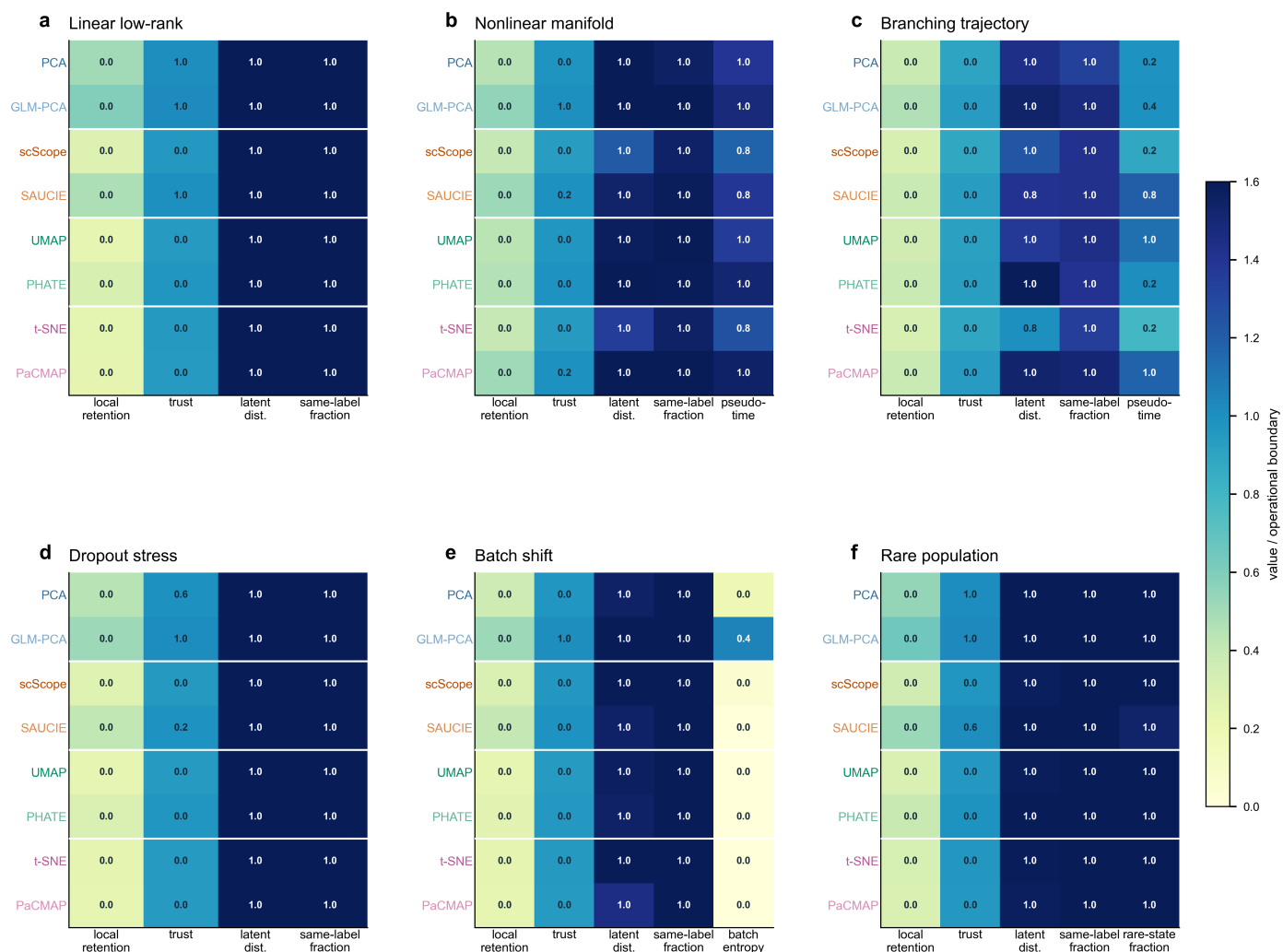


Supplementary Figures

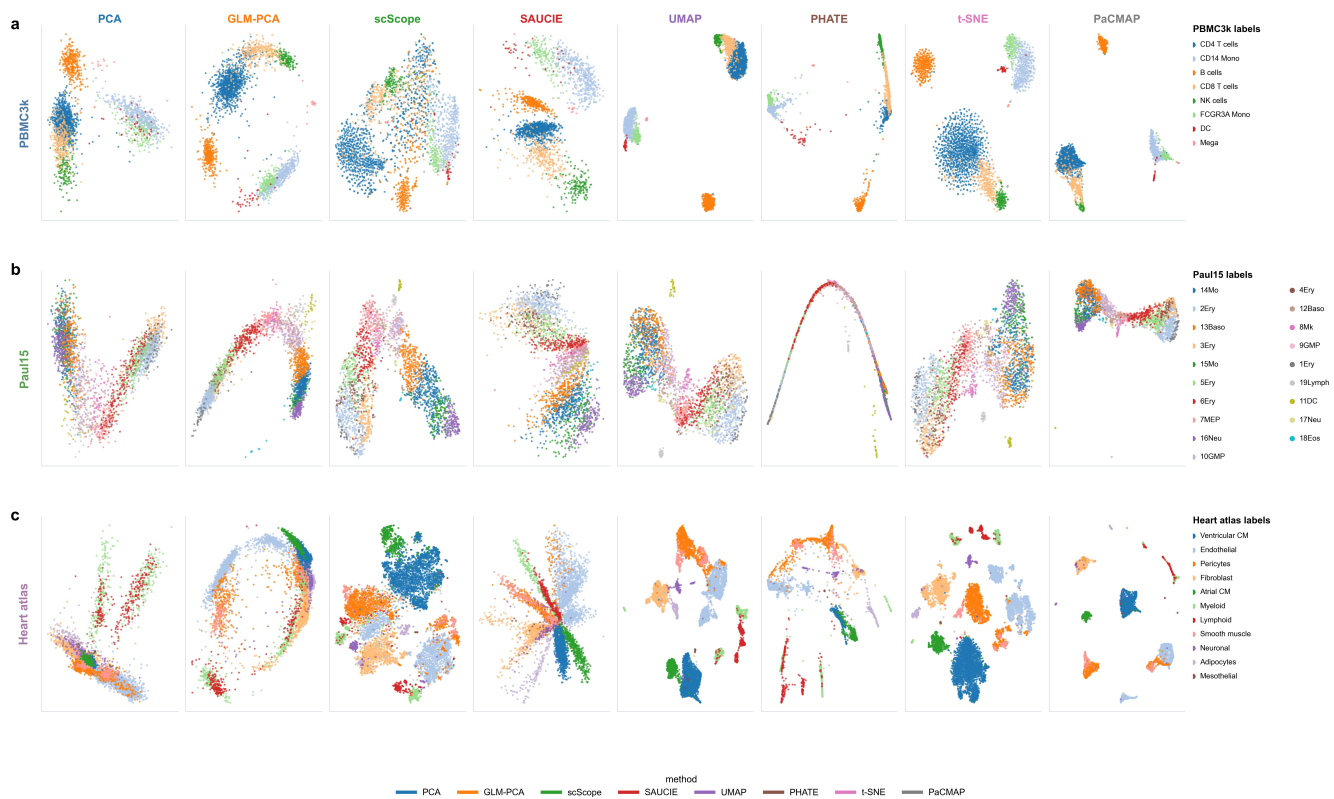
Mathematical characterization and empirical evaluation of single-cell transcriptomic
dimensionality-reduction methods



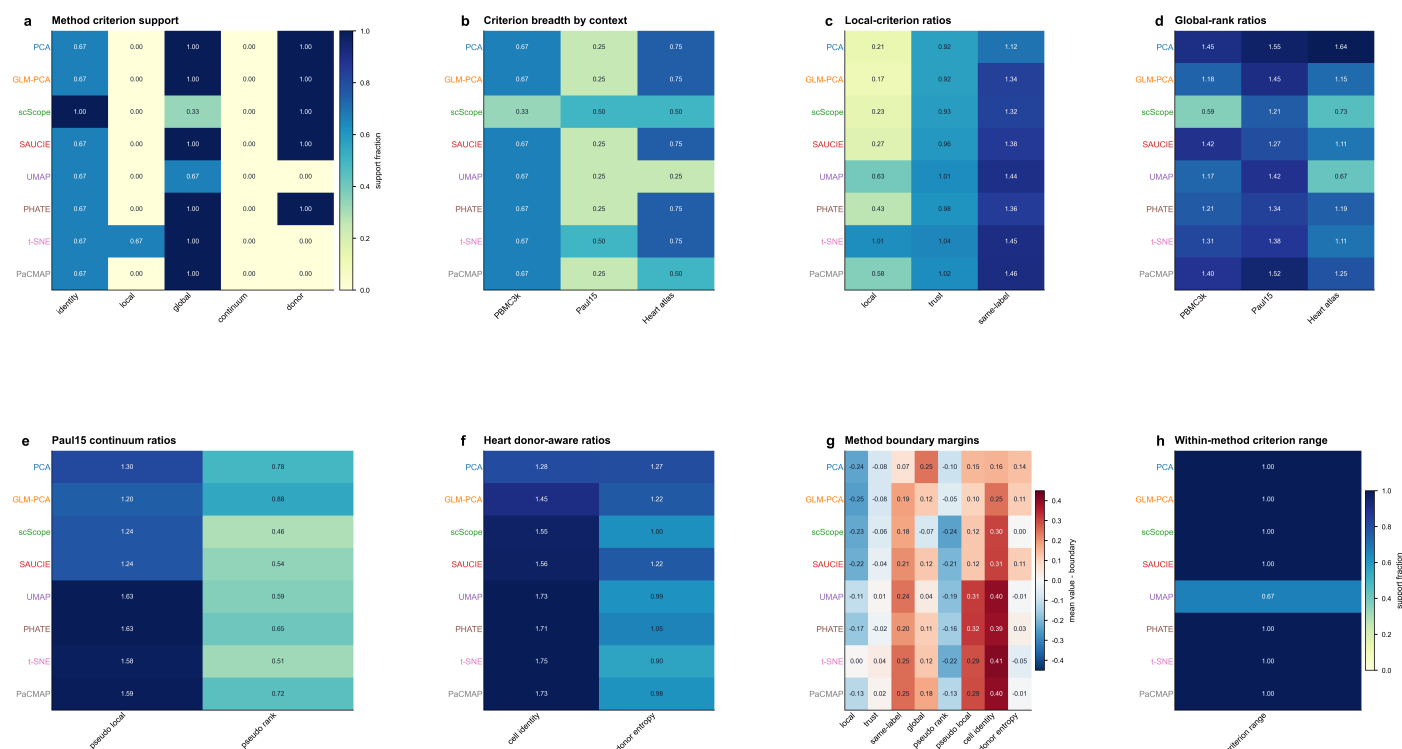
Supplementary Fig. S1. Dataset structure and analysis context for property-specific embedding interpretation. **a**, Dataset scale for PBMC3k, Paul15 and the heart-atlas subset. Points show cells and genes in each processed analysis object; point area denotes the number of annotation levels used in the diagnostic analyses. **b**, Label-size distributions. Each point is one annotation level and the dashed line marks the fixed 200-cell rare-state boundary. **c**, Donor-level cell counts in the heart-atlas subset. **d**, Label-burden metrics, comprising class imbalance, largest-label fraction, rare-cell fraction, rare-label fraction and smallest-label fraction. **e**, Expression sparsity, shown as zero fraction against mean detected highly variable genes per cell; point area denotes cell count. **f**, Cumulative variance explained at PC5, PC10 and PC20. **g**, PCA-reference separability, measured by PC-space same-label neighbour fraction and the PC20 silhouette coefficient. **h**, Detection-depth spread; points are medians and whiskers span the 10th to 90th percentiles of detected highly variable genes per cell. **i**, Median and mean total counts per cell. **j**, Rare-state depth burden, combining rare-cell fraction, rare-label fraction and scaled median detected genes; text gives the number of rare annotation levels. These are empirical dataset summaries rather than method-performance results. Here, n denotes datasets, annotation levels, donors or cells as specified by the panel. Public dataset sources, analysed objects, diagnostic criteria and biological definitions are reported in Supplementary Tables S1, S2, S4 and S5; figure-level numerical records and plotting code are available in the public repository.



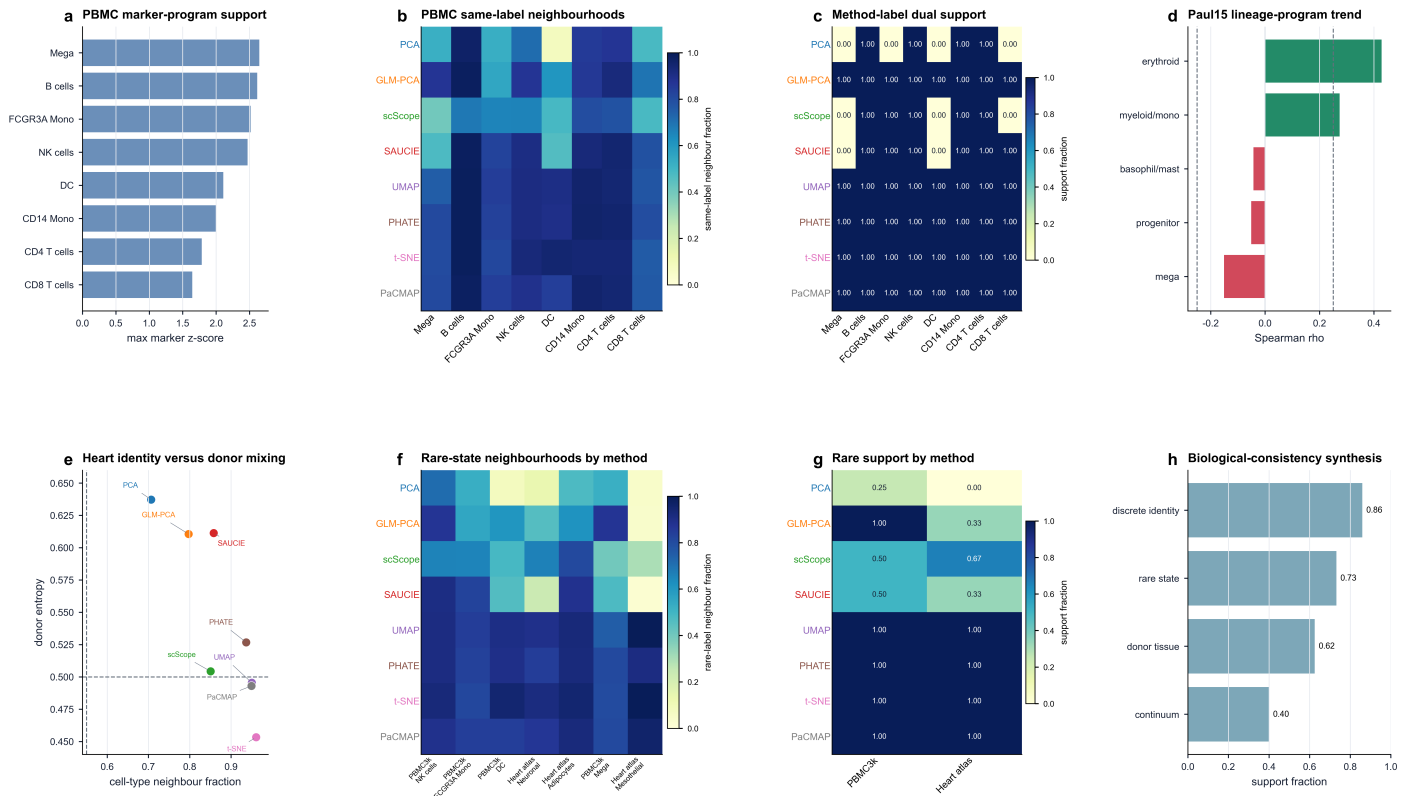
Supplementary Fig. S2. Metric-level summary of the eight-method controlled stress-test suite. a-f, Scenarios with known generating structure designed to examine property-specific vulnerabilities. Panels show linear low-rank structure, nonlinear manifold structure, branching trajectory, dropout stress, batch shift and rare-population scenarios. Rows are PCA, GLM-PCA, scScope, SAUCIE, UMAP, PHATE, t-SNE and PaCMAP. Columns show scenario-relevant diagnostics, including local retention against the generating structure, trustworthiness, latent-distance correlation, same-label neighbour fraction, pseudotime-distance correlation, batch entropy and rare-label neighbour fraction. Only metrics defined and calculated for a scenario are displayed. Colour encodes the observed value divided by its fixed operational boundary, capped at 1.6 for display; in-cell numbers give the fraction of five independently generated datasets meeting the boundary. These panels are controlled computational results and do not represent new empirical biology or a comprehensive method benchmark. Operational criteria are reported in Supplementary Table S4, and complete method-scenario-generated-dataset results are reported in Supplementary Table S13.



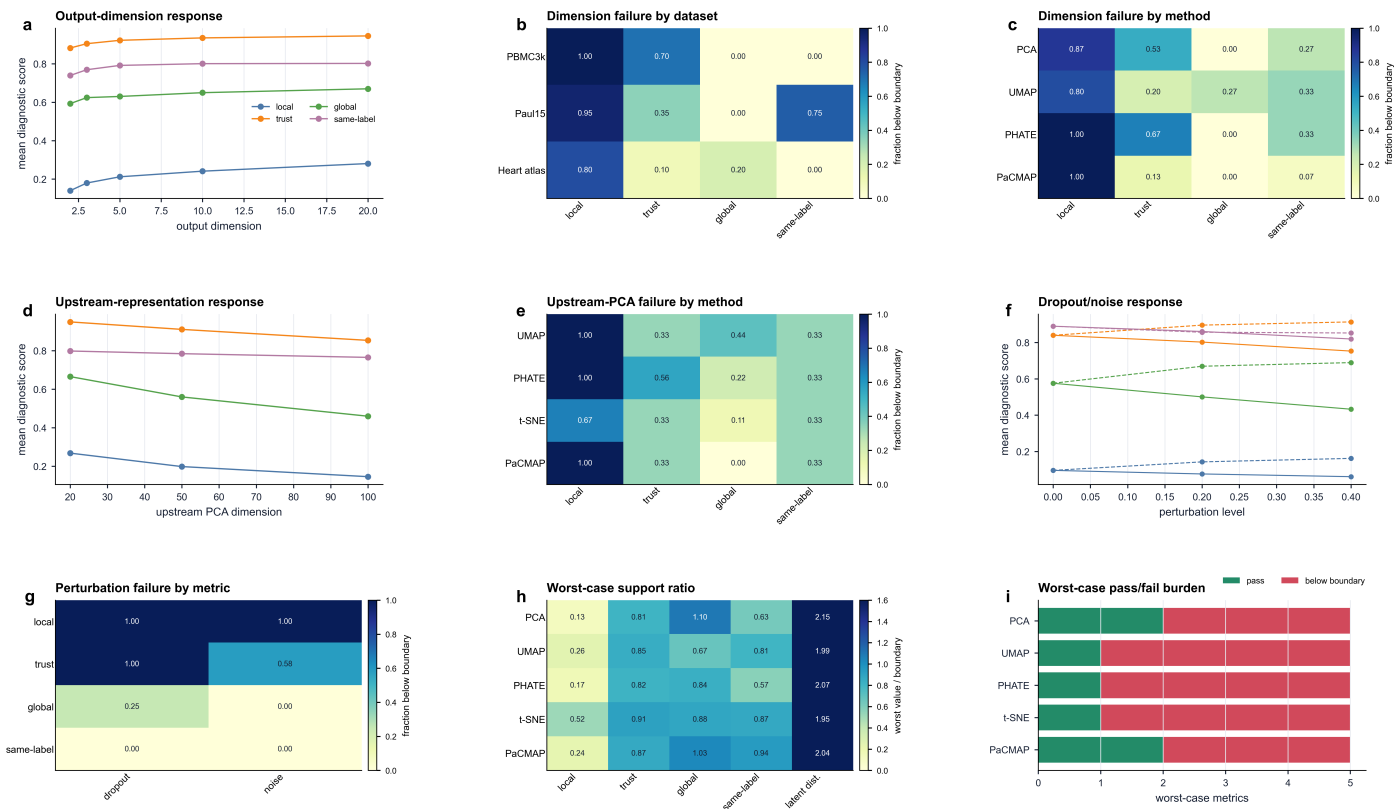
Supplementary Fig. S3. Full embedding atlas across empirical interpretation contexts. a-c, Low-dimensional coordinates for the complete eight-method validation panel in PBMC3k, Paul15 and the heart-atlas subset. Columns show PCA, GLM-PCA, scScope, SAUCIE, UMAP, PHATE, t-SNE and PaCMAP in the same fixed order used throughout the study. Points are cells and colours denote the dataset-specific annotation labels shown at the right of each row. Method-name colours are navigational identifiers only and do not encode a method class or performance level. These maps document the fitted coordinates underlying the matched comparisons; visual separation is not used as a quantitative result. Diagnostics and biological-consistency checks are reported in Figs 3-5. Here, n is the number of cells in the processed analysis object for each dataset-method map; fitted coordinates and plotting code are available in the public repository.



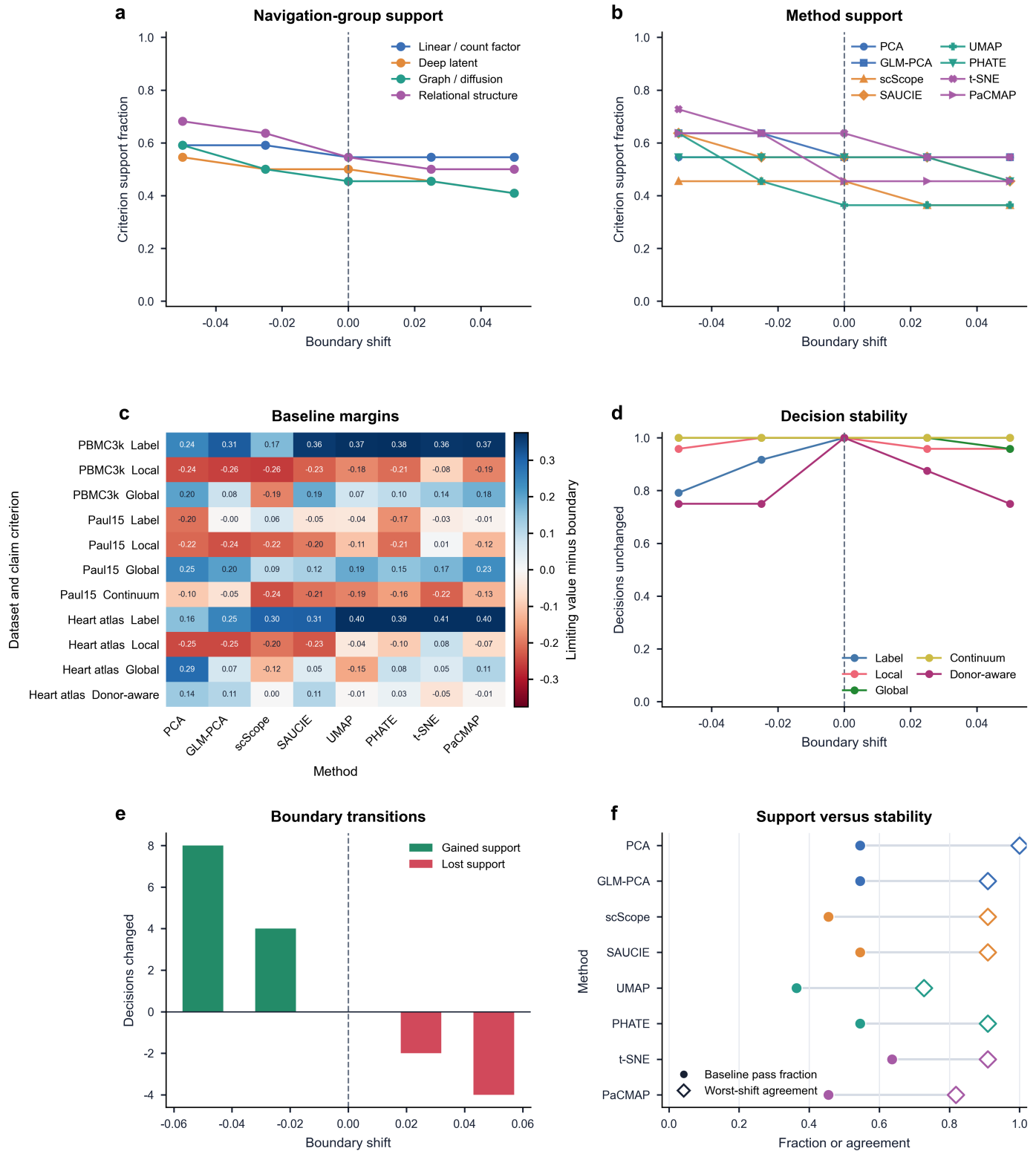
Supplementary Fig. S4. Diagnostic support, component ratios and boundary margins. **a**, Method-level support fractions for label, local-neighbourhood, global-geometry, continuum and donor-aware criteria. **b**, Fraction of applicable criteria supported by each method in PBMC3k, Paul15 and the heart-atlas subset. **c**, Local-criterion component ratios, calculated as the observed value divided by its fixed operational boundary for local retention, trustworthiness and same-label neighbour fraction. **d**, Global-rank ratios across the three empirical contexts. **e**, Paul15 continuum ratios for local pseudotime retention and pseudotime-rank correlation. **f**, Heart-atlas donor-aware ratios for cell-type neighbour fraction and normalised donor entropy. **g**, Mean signed boundary margin for each method and diagnostic component, calculated as observed value minus its boundary. **h**, Within-method criterion range, calculated as the maximum minus minimum valid interpretation-specific support fraction. Heatmap colour denotes support fraction, value-to-boundary ratio or signed margin as specified by each colour bar. Method-name colours identify methods only. Fractions use valid method-context or method-interpretation rows as denominators; unavailable criteria are excluded rather than counted as failures. Diagnostic definitions and boundaries are reported in Supplementary Table S4; panel-level data and plotting code are available in the public repository.



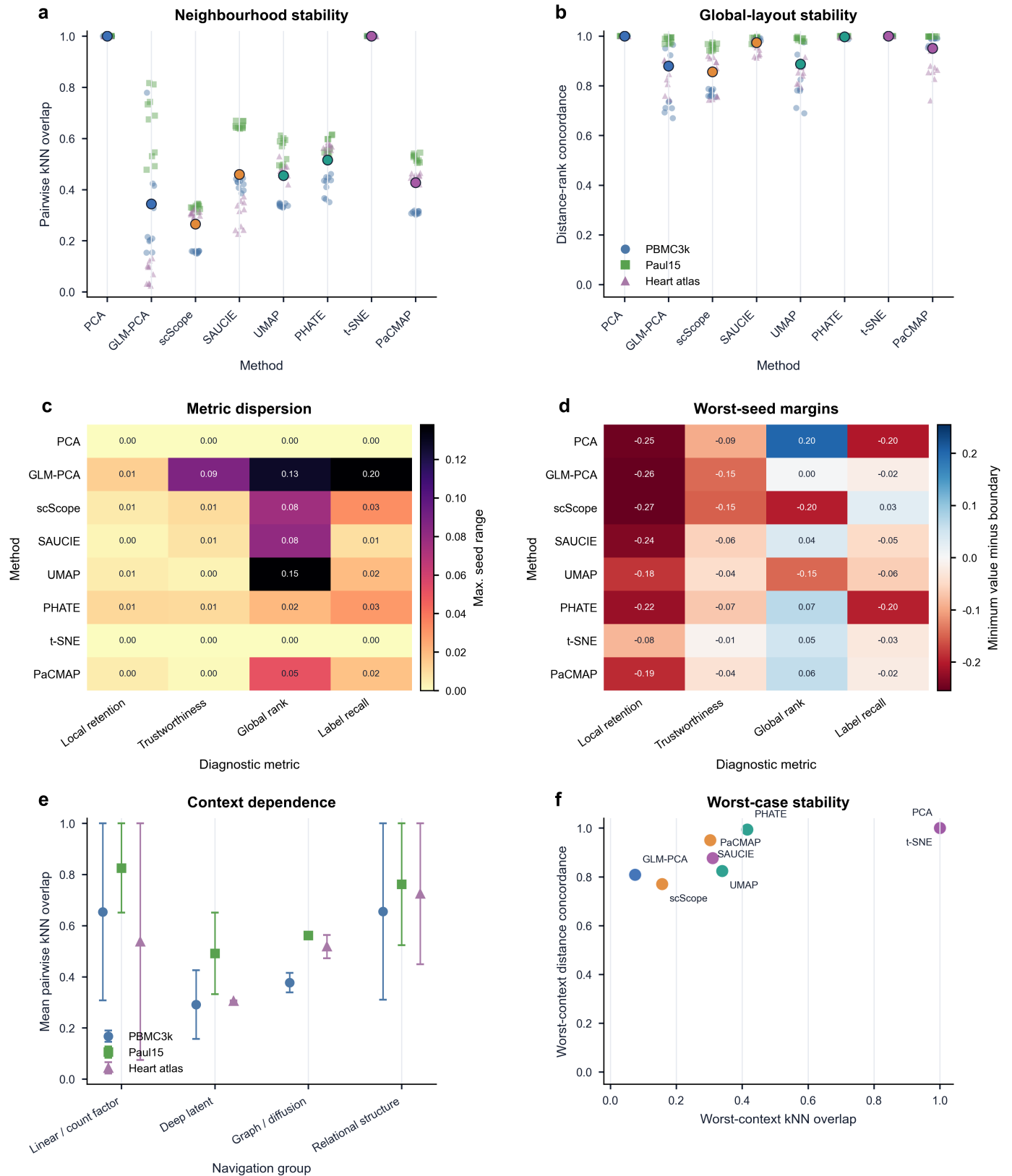
Supplementary Fig. S5. Expression- and metadata-based consistency checks for embedding interpretation. **a**, Maximum marker-program z score for each PBMC3k annotation. **b**, PBMC3k same-label neighbour fraction by method and label. **c**, Binary method-label support requiring both the marker-program and neighbour criteria. **d**, Spearman correlation between Paul15 lineage-program score and reference pseudotime; dashed lines mark the fixed absolute-correlation boundary of 0.25. **e**, Heart-atlas cell-type neighbour fraction plotted against normalised donor entropy; dashed lines mark the corresponding diagnostic boundaries. **f**, Rare-label neighbour fraction by method for four PBMC3k and three heart-atlas low-frequency labels. **g**, Fraction of rare labels meeting the neighbour-fraction boundary for each method and dataset. **h**, Descriptive support fractions across discrete identity, rare-state, donor-variable tissue and continuum checks. Method-name colours are identifiers only. Marker programs, pseudotime-associated expression trends and donor metadata were defined without using the displayed embedding coordinates, but they are consistency checks rather than independent experimental validation. Here, n denotes method-label, method-rare-label, lineage-program or method rows. Diagnostic and biological definitions are reported in Supplementary Tables S4 and S5; panel-level data and plotting code are available in the public repository.



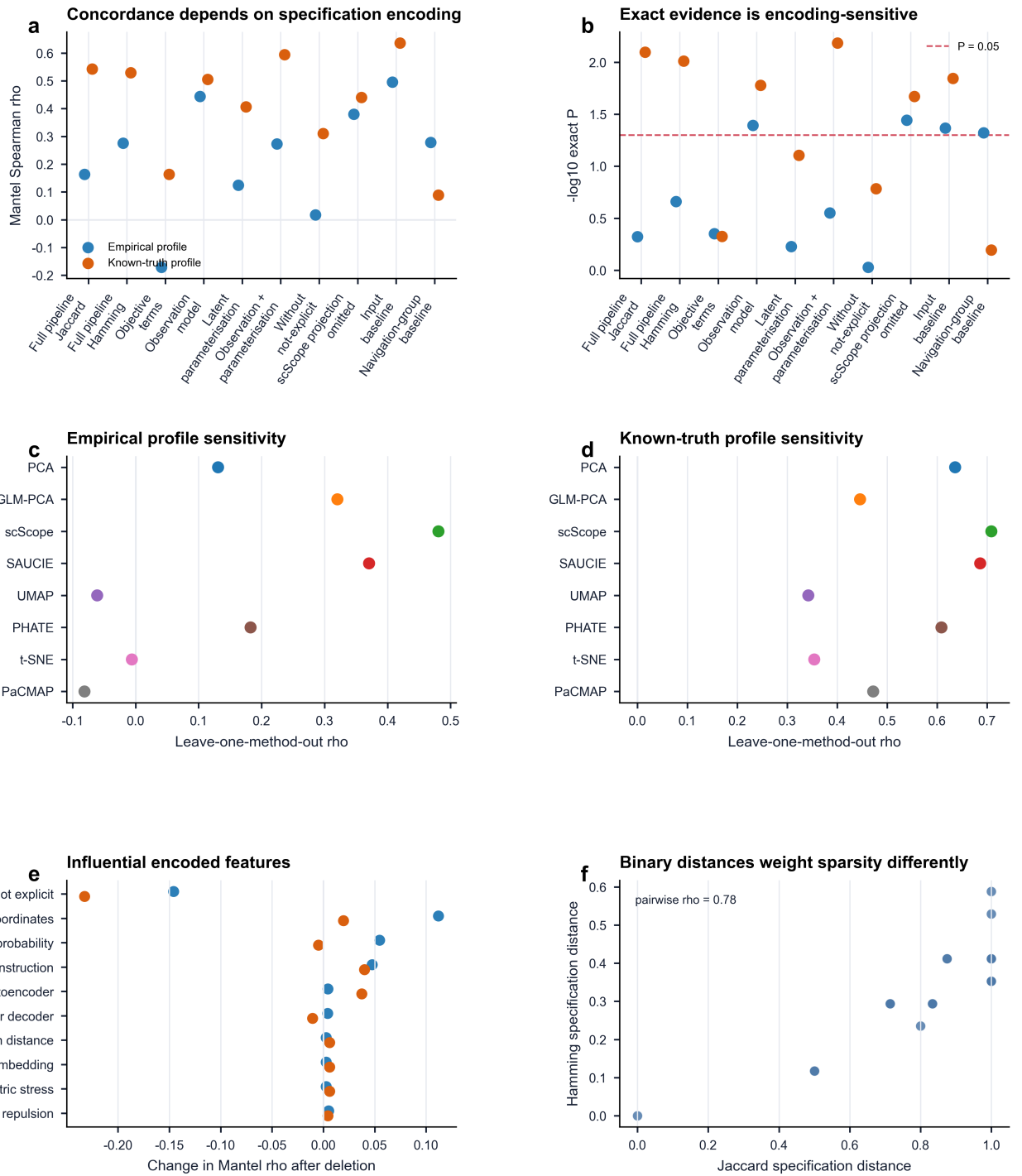
Supplementary Fig. S6. Robustness responses across dimensionality, upstream representation and perturbation. **a**, Mean diagnostic response across output dimensions for local retention, trustworthiness, global-rank correlation and same-label neighbour fraction. **b**, Failure fractions under output-dimension variation by dataset and endpoint. **c**, Failure fractions under output-dimension variation by recomputed method and endpoint. **d**, Mean diagnostic response across upstream PCA dimensions for UMAP, PHATE, t-SNE and PaCMAP. **e**, Failure fractions under upstream-PCA variation by method and endpoint. **f**, Mean diagnostic response under dropout and noise perturbations; solid lines denote dropout and dashed lines denote noise. **g**, Perturbation failure fraction by endpoint and perturbation type. **h**, Worst-case support ratio for the recomputable subset, calculated as the minimum observed value divided by its fixed operational boundary across the tested conditions. **i**, Worst-case pass and below-boundary burden by method for the same subset. Method colours identify methods and shaded ranges show observed minima and maxima rather than confidence intervals. Panels summarise computational sensitivity analyses, not independent biological replicates. Diagnostic criteria and complete matched robustness records are reported in Supplementary Tables S4 and S12; panel-level data and plotting code are available in the public repository.



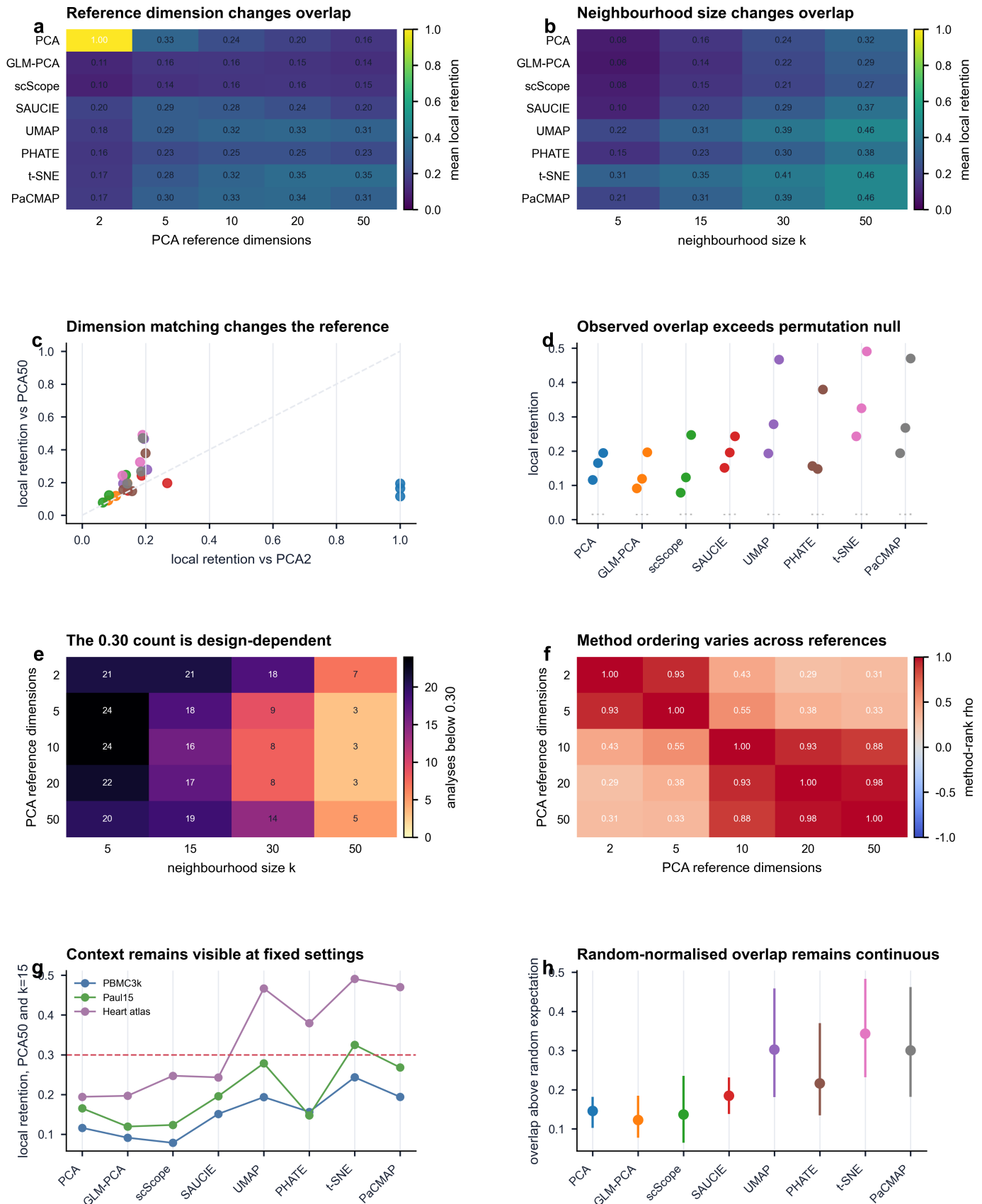
Supplementary Fig. S7. Interpretation criteria are tested across a defined range of operational boundaries. **a**, Fraction of applicable decisions meeting the shifted boundary within each method group. Each point summarises 22 method-context decisions from the two empirical-panel methods in that group. **b**, Criterion pass fraction for each method across its 11 applicable dataset-interpretation combinations. Colours identify method groups and symbols identify methods. **c**, Baseline limiting margin for all 88 method-context decisions. Rows are the 11 applicable dataset-interpretation combinations and columns are methods. Each cell is the minimum observed-minus-boundary value among the required components; red denotes a limiting component below its boundary and blue denotes a value above it. **d**, Fraction of decisions unchanged from the unshifted result for label support, local neighbourhoods, global geometry, continuum and donor-aware interpretation. Denominators are 24, 24, 24, 8 and 8 decisions, respectively. **e**, Number of decisions that gained or lost support relative to baseline among all 88 decisions. **f**, Method-level baseline pass fraction and worst-shift decision agreement. Filled circles show the baseline fraction over 11 applicable decisions, open diamonds show the minimum unchanged fraction across the four non-zero shifts and connecting lines aid within-method comparison. Every continuous component boundary was shifted jointly by -0.05, -0.025, 0, 0.025 or 0.05; the vertical dashed line denotes the baseline. These are deterministic sensitivity calculations, not independent replicates, and no confidence intervals or inferential P values are assigned. Component values, boundaries, decisions and denominators are reported in Supplementary Tables S4 and S7.



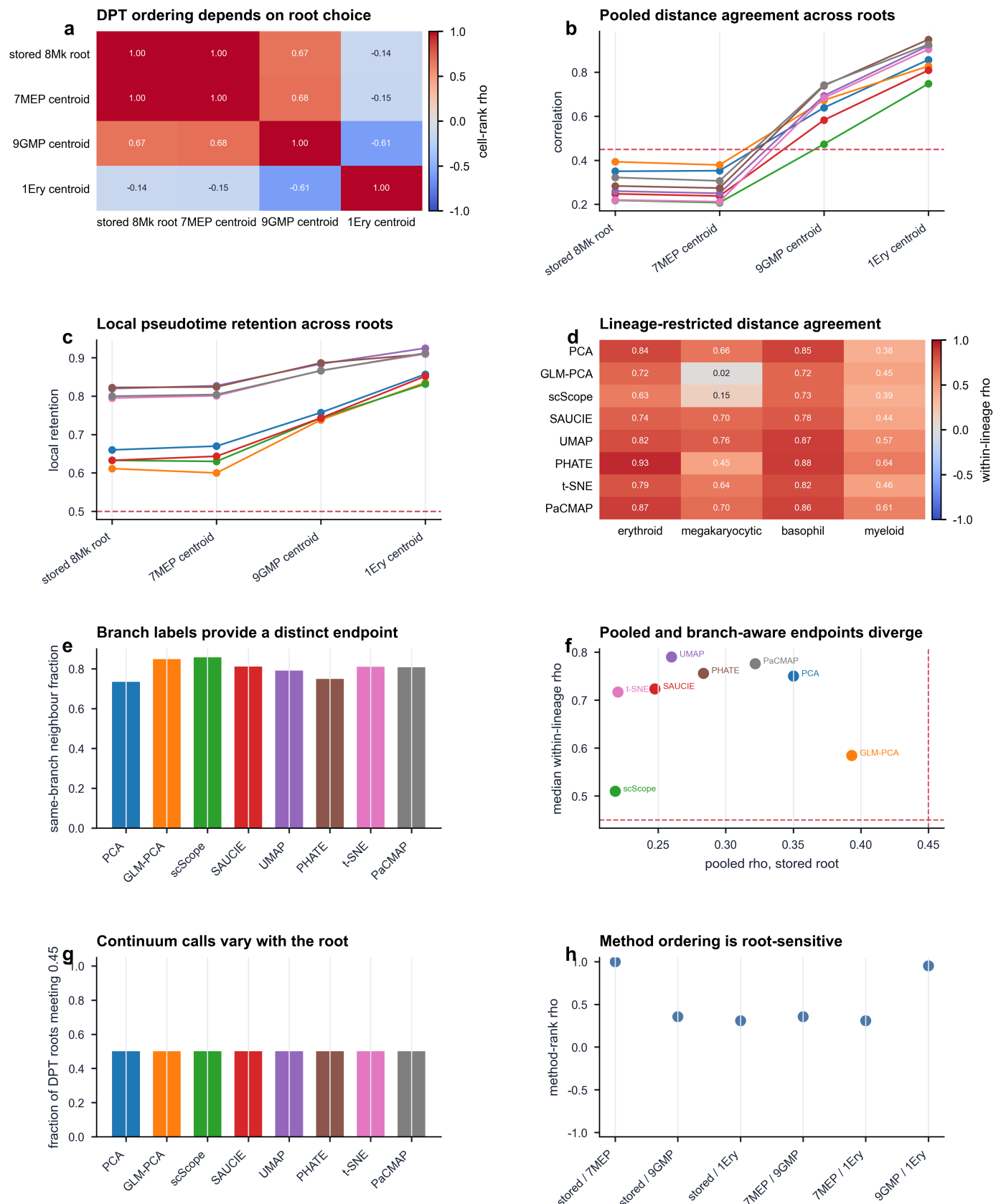
Supplementary Fig. S8. Local and global embedding structures show distinct random-seed stability. All eight methods were fitted and scored on the complete PBMC3k, Paul15 and heart-atlas analysis objects, containing 2,638, 2,730 and 8,000 cells, with seeds 0–4. **a**, Pairwise k-nearest-neighbour overlap at $k=15$. Small points are the ten seed-pair comparisons for each dataset and method; circles, squares and triangles denote PBMC3k, Paul15 and heart atlas. Large outlined circles show the mean across 30 seed-pair comparisons per method, with colour identifying its navigation group. **b**, Pairwise Spearman concordance of embedding-distance ranks over 5,000 fixed cell pairs, using the same point and colour conventions. **c**, Maximum diagnostic range across five seeds, taking the largest range among the three datasets for each method. **d**, Worst seed and context boundary margin, calculated as the minimum observed-minus-boundary value over 15 method-dataset-seed runs. **e**, Context-specific mean pairwise neighbourhood overlap by navigation group. Points average the two methods in each group and whiskers span their observed minimum and maximum; whiskers are descriptive ranges, not confidence intervals. **f**, Worst-context stability for each method. Coordinates are the minima of the three context-specific means for neighbourhood overlap and distance-rank concordance. PCA and the evaluated Scanpy t-SNE configuration produced identical coordinates across five seeds in the recorded software stack; this execution-specific result does not imply general determinism. Seed pairs are computational repeats on the same cells and are not biological replicates. Run-level metrics, comparisons, decisions and software settings are reported in Supplementary Tables S6 and S8.



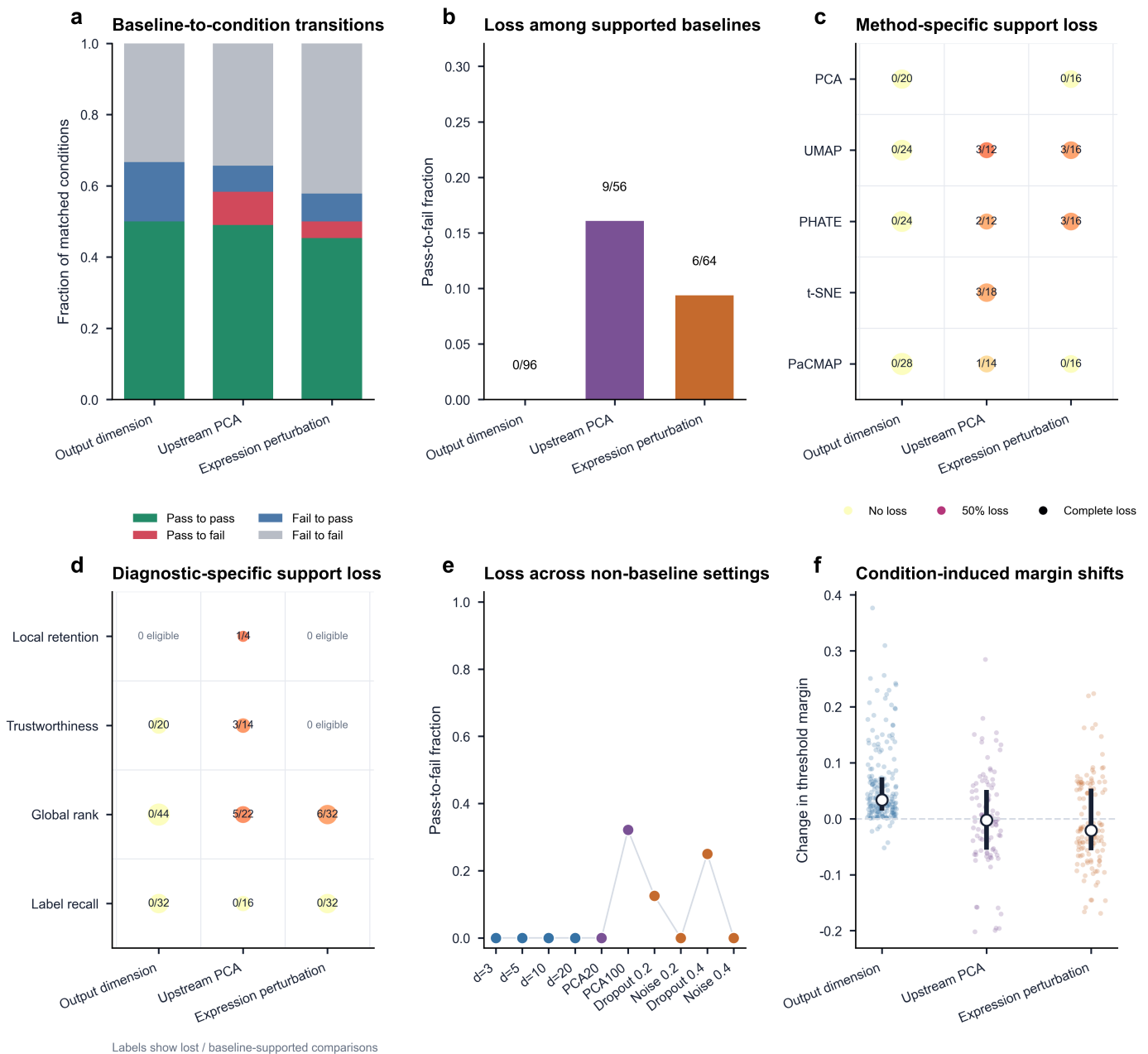
Supplementary Fig. S9. Specification-to-profile concordance depends on encoding and method composition. **a**, Exact Mantel Spearman correlation between ten alternative specification encodings and the empirical or controlled-scenario profile. Each point uses eight methods and 28 unique method pairs. **b**, Negative base-10 logarithm of the corresponding exact two-sided P value; the dashed line marks unadjusted $P = 0.05$. Alternative encodings are exploratory and were not selected by these P values. **c**, **d**, Leave-one-method-out Mantel correlations for empirical and controlled-scenario profiles. Each point uses seven methods, 21 pairs and all 5,040 method-label permutations. **e**, Change in Mantel correlation after deleting one encoded feature from the full specification; the ten features with the largest absolute change across the two profiles are shown. **f**, Jaccard against Hamming distance for the 28 method pairs under the same 17-feature encoding. The displayed Spearman correlation describes the pairwise relation and is not an independent-pair significance test. Blue and orange identify empirical and controlled-scenario profiles. Complete encodings and exact permutation results are reported in Supplementary Table S9.



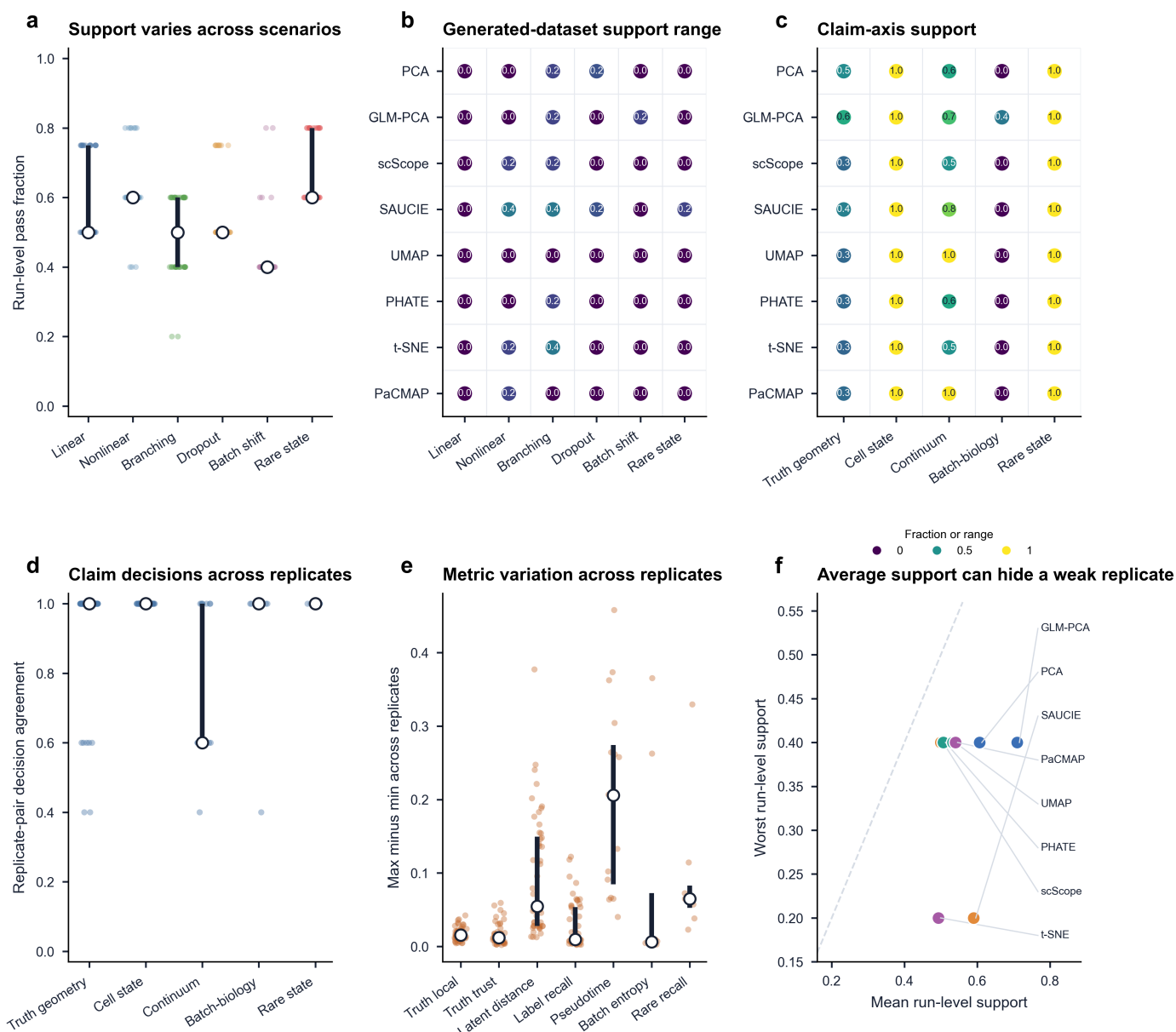
Supplementary Fig. S10. Local-retention conclusions depend on the reference dimension and neighbourhood size. A fixed label-stratified subset of 1,000 cells per dataset was used for every calibration. a, Mean local retention by method across PCA reference dimensions 2, 5, 10, 20 and 50 at fixed k. b, Mean local retention across neighbourhood sizes 5, 15, 30 and 50 at fixed reference dimension. c, Local retention under PCA2 against PCA50 for matched method-context evaluations; the diagonal marks equality. d, Observed local retention at the primary calibration setting against an unrestricted cell-correspondence permutation null. Coloured points are datasets within methods and grey marks show null expectations. e, Number of 24 method-context evaluations below 0.30 for all 20 reference-dimension-by-k definitions. f, Spearman concordance of eight-method ranks between PCA reference dimensions. g, Context-resolved local retention at PCA50 and k = 15; the dashed line marks 0.30. h, Observed overlap above random expectation by method; points are context means and whiskers span the observed context range. These calibration results are distinct from the full-object primary analysis. Complete values and permutation definitions are reported in Supplementary Table S10.



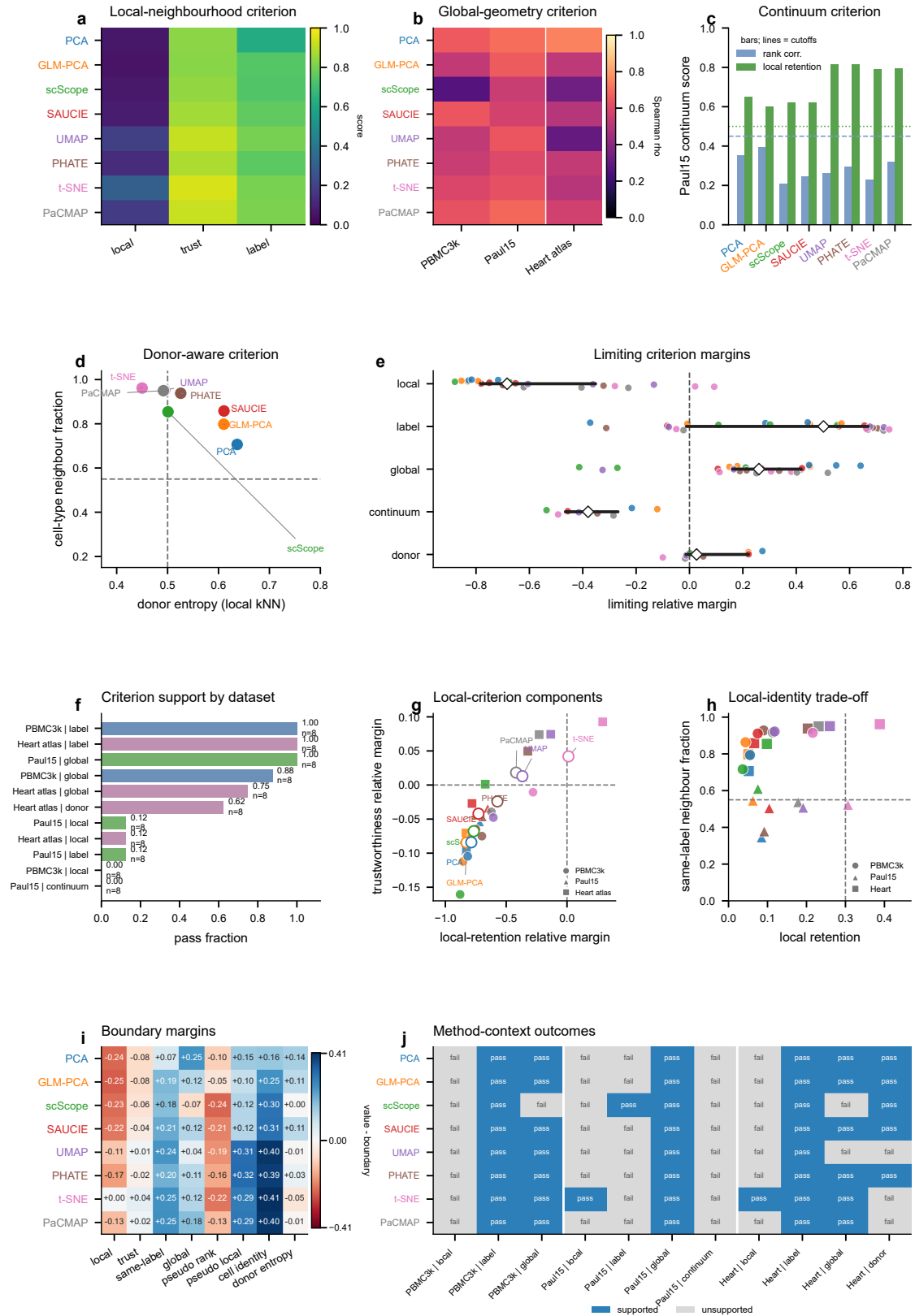
Supplementary Fig. S11. Paul15 continuum diagnostics are sensitive to diffusion-pseudotime root and lineage. All analyses use the same 2,730-cell Paul15 object. **a**, Spearman correlation of cell pseudotime ranks between the stored 8Mk root and 7MEP, 9GMP or 1Ery centroid roots. **b**, Pooled pseudotime-distance correlation for eight methods under each root; the dashed line marks 0.45. **c**, Local pseudotime retention under the same roots; the dashed line marks 0.50. **d**, Within-lineage pseudotime-distance correlation for erythroid, megakaryocytic, basophil and myeloid cluster sets. **e**, Same-branch neighbour fraction by method. **f**, Stored-root pooled correlation against the median lineage-restricted correlation; dashed lines mark 0.45. **g**, Fraction of four roots for which pooled pseudotime-distance correlation met 0.45. **h**, Spearman concordance of eight-method ranks between root pairs. Root choices are alternative analyses of one object and are not independent replicates. Lineage cell counts, sampled-pair denominators and root definitions are reported in Supplementary Table S11.



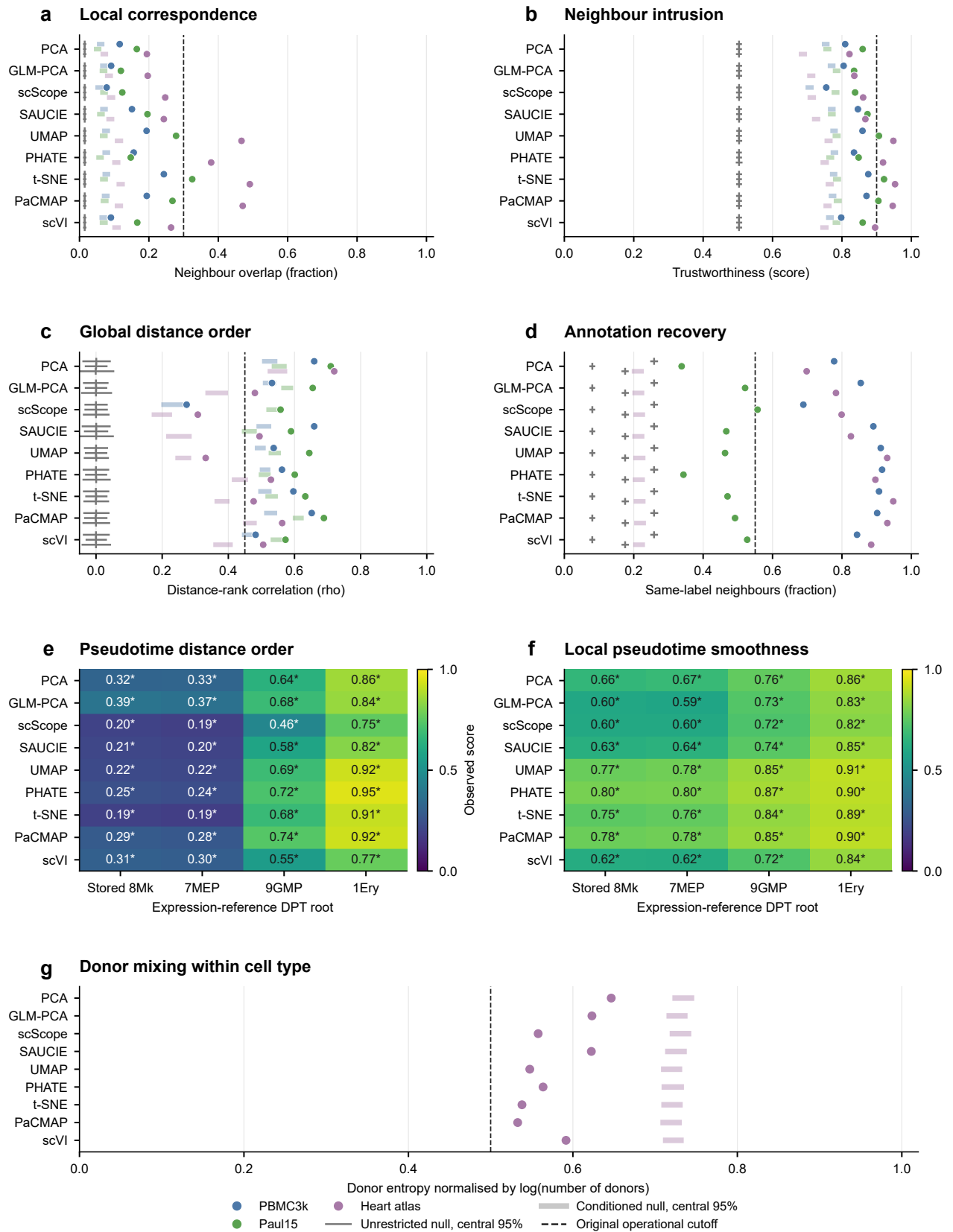
Supplementary Fig. S12. Matched robustness transitions distinguish unsupported baselines from perturbation-induced support loss. **a**, Fractions of matched comparisons classified as pass-to-pass, pass-to-fail, fail-to-pass or fail-to-fail for output dimension, upstream PCA and expression perturbation. **b**, Pass-to-fail fraction among baseline-supported comparisons; labels give lost over eligible comparisons. **c**, Method-specific support loss. Point area and colour encode the loss fraction, and labels give lost over baseline-supported comparisons; blank cells are method-axis combinations not evaluated. **d**, Diagnostic-specific support loss using the same encoding. Text marked 0 eligible indicates that no baseline comparison met the boundary. **e**, Pass-to-fail fraction for each non-baseline setting. **f**, Change in signed boundary margin from baseline to condition. Small points are matched comparisons; open circles are medians and vertical bars span the interquartile range. Comparisons are computational conditions on the same analysis objects, not biological replicates. Transition counts and denominators are reported in Supplementary Table S12.



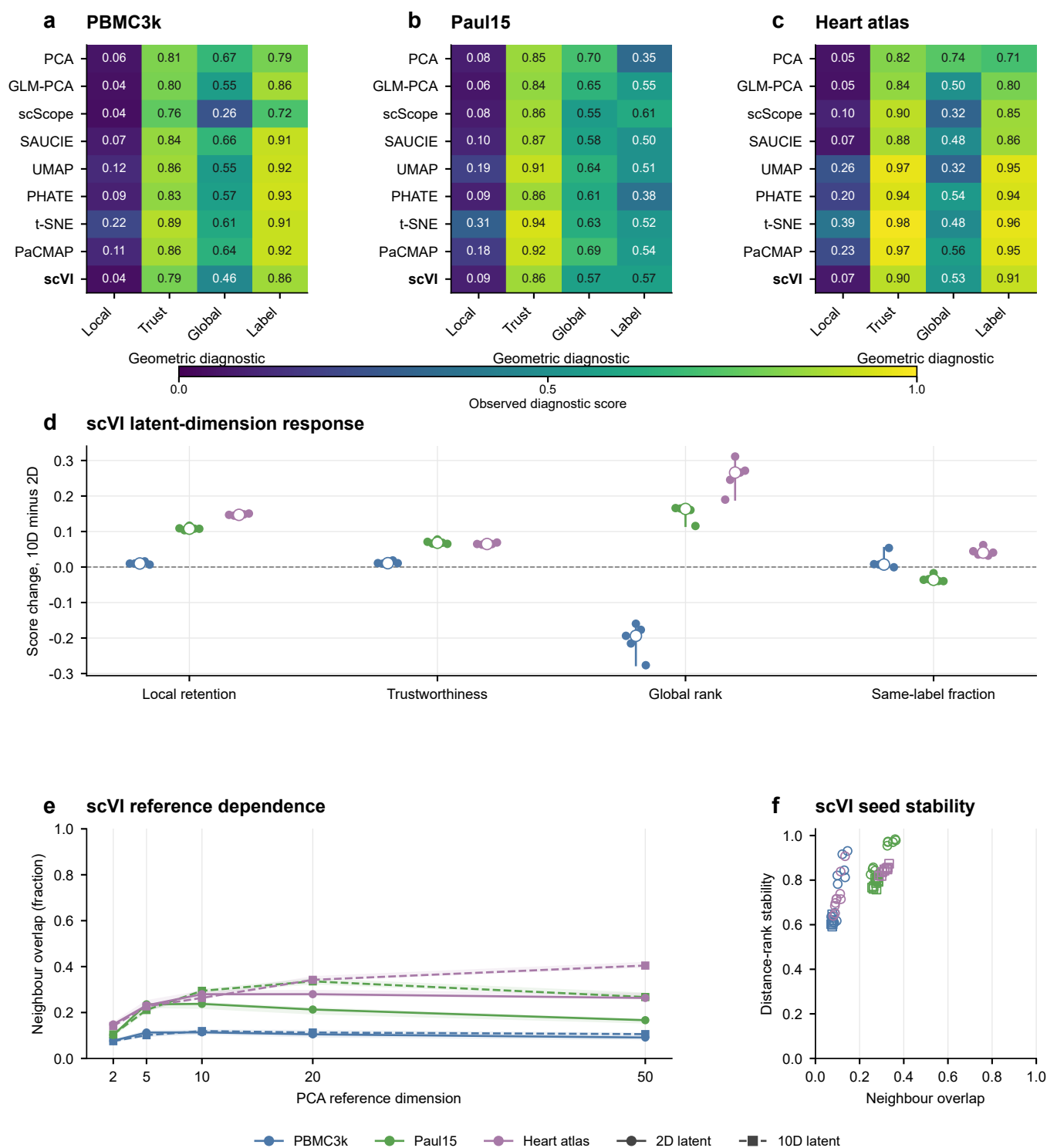
Supplementary Fig. S13. Five independently generated datasets per scenario quantify controlled stress-test stability. **a**, Run-level support fraction across 40 runs for each scenario, comprising eight methods and five generated datasets. Small points are runs; open circles and vertical bars summarise the distribution. **b**, Range in run-level support across the five generated datasets for every method-scenario combination. In-cell values give the range from 0 to 1. **c**, Interpretation-axis pass fraction for every method across all applicable generated-dataset metric rows. Point colour and label encode the fraction. **d**, Pairwise decision agreement among the five generated datasets for generating-structure geometry, cell state, continuum, batch-biology trade-off and rare-state interpretations. **e**, Maximum-minus-minimum metric value across generated datasets for each method-scenario-metric combination; points are combinations and open circles with bars summarise distributions. **f**, Mean run-level support against worst-run support for each method. The dashed diagonal marks equality, and connectors associate displaced labels with points. Generated datasets are computational replications rather than biological replicates. The analysis contains 30 generated datasets, 240 complete runs and 1,120 valid metric rows; complete results are reported in Supplementary Table S13.



Supplementary Fig. S14. Operational diagnostic criteria and threshold sensitivity in the eight-method anchor panel. **a**, Mean local retention, trustworthiness and same-label neighbour fraction across PBMC3k, Paul15 and the heart subset for each method. **b**, Global-rank correlation by method and dataset. **c**, Paul15 pseudotime-distance correlation and local pseudotime retention under the stored 8Mk-root definition; dashed lines mark operational boundaries of 0.45 and 0.50. **d**, Heart-subset donor entropy and same-label-type-neighbour fraction; dashed lines mark operational boundaries of 0.50 and 0.55, and the grey connector associates the displaced scScope label with its point. **e**, Limiting signed relative margin for each applicable method-context-claim decision, defined as the smallest component value of $(\text{value} - \text{boundary}) / \text{boundary}$ within a criterion. Coloured points are decisions, black segments span the interquartile range, open diamonds are medians and the dashed line marks zero. **f**, Criterion support by dataset and claim; labels report the supported fraction and number of evaluated method rows. **g**, Joint local-criterion component plane for 24 method-context evaluations. Solid points are individual evaluations, open circles are method means, colour identifies method and symbol identifies dataset. **h**, Local retention against same-label neighbour fraction for the same 24 runs. **i**, Mean observed-minus-boundary margin for each method and diagnostic component. **j**, Binary outcomes for all applicable method-context-claim criteria; blank combinations were not defined and were not counted as failures. The boundaries are study-level operational decision rules, not statistical significance thresholds or universal performance standards. Continuous values and margins are the primary evidence; the binary summaries are retained for sensitivity and auditability.



Supplementary Fig. S15. Random-reference calibration distinguishes non-random structure from operational fidelity. a-d, Local neighbour overlap, trustworthiness, global distance-rank correlation and same-label neighbour fraction for nine methods on matched 1,000-cell subsets of PBMC3k, Paul15 and heart atlas. Blue, green and purple points identify datasets. Grey bars span central 95% intervals from 1,999 unrestricted permutations; grey ticks mark null means. Coloured translucent bars show annotation-conditioned geometry nulls in a-c and donor-conditioned label permutations for heart in d. Dashed lines mark unchanged operational cutoffs of 0.30, 0.90, 0.45 and 0.55. Local overlap is the shared fraction of 15 nearest neighbours; trustworthiness penalises embedding-only neighbours; global ρ compares distance ranks across 5,000 non-self pairs; label fraction measures annotation agreement. e, f, Observed pseudotime-distance ρ and local pseudotime retention under four expression-reference diffusion pseudotime roots. 8Mk, 7MEP, 9GMP and 1Ery denote megakaryocyte, megakaryocyte-erythroid progenitor, granulocyte-monocyte progenitor and erythroid annotations. Retention is one minus neighbour-to-random-pair mean pseudotime difference. Numbers and colour show observed scores; asterisks denote upper-tail $q < 0.05$ relative to coarse-lineage-conditioned permutations. g, Normalised heart donor entropy, with observed points and central 95% intervals from donor permutations within cell types. All nine lower-tail deficit tests have $q = 0.0005$. Donor deficits identify donor association, not technical-batch causality. Probabilities use the plus-one Monte Carlo estimator and separate Benjamini-Hochberg adjustment within each null family. Null intervals are not confidence intervals; cells, pairs and permutations are not biological replicates. scVI is unsupervised and donor-unconditioned. Full effects, probabilities, denominators and cutoff decisions are reported in Supplementary Table S15.



Supplementary Fig. S16. scVI geometry depends on biological context and reference definition. a-c, Four geometric diagnostics for the eight anchor methods and scVI at output dimension two and seed zero in PBMC3k, Paul15 and heart atlas. Local, Trust, Global and Label denote reference-neighbour overlap, trustworthiness, Spearman distance-rank correlation and same-label neighbour fraction. Colour and numbers show observed scores; the common scale is zero to one. The full objects contain 2,638, 2,730 and 8,000 cells. Neighbour size is 15, the reference is PCA50 and global scores use 5,000 fixed non-self cell pairs with sampling seed 20261005. d, Paired score changes between ten- and two-dimensional scVI latents. Small points denote five seeds, open circles medians and vertical bars observed ranges; the horizontal line marks no change. e, Local overlap across reference PCA dimensions on matched 1,000-cell subsets. Lines show medians over five seeds and shading their observed ranges. Solid circles denote two-dimensional latents and dashed squares ten-dimensional latents. f, Neighbour overlap against distance-rank agreement between independently seeded scVI outputs. Each context and dimension contributes ten seed-pair comparisons formed from five seeds; circles denote two-dimensional and squares ten-dimensional latents. Blue, green and purple denote PBMC3k, Paul15 and heart atlas throughout d-f. scVI was fitted without annotations or donor covariates, so these panels do not evaluate batch-conditioned scVI. Ranges describe computational variation, not confidence intervals. Cells, sampled pairs, seeds and shared seed pairs are not independent biological replicates. No inferential tests or significance symbols are used. Full diagnostic, reference-sensitivity and stability values are reported in Supplementary Table S16.